Recommended Assessment

Thermocouple

Exploration

1. When manually warming up the probe, what is the minimum temperature change that is read by the thermocouple probe? How does this compare to what the datasheet shows?

2. After testing with a voltmeter, what happened when you checked for continuity between the two connectors? When you tested with one of the leads and the metal casing, what did you notice? What type of junction did the thermocouple have?

Testing the Thermocouple

3. The datasheet mentions the following: "While the temperature at the cold end fluctuates, the device continues to accurately sense the temperature difference at the opposite end." Describe what you observed when testing this and add the corresponding screenshot.

4. Based on your results from measuring the temperature in a hot or cold liquid, analyze how long it takes to reach the final stable temperature value. Calculate the time constant, which is how long it takes to reach 63.2% of the final temperature? Use the screenshot of your plot.

5. Did you observe any difference in the measurement based on how the probe was placed on the liquid to measure temperature? If so, describe your observations.

Thermocouple Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	